**Week 9: 26/10 – 30/10**

**Subject: Math - Statistics** **Unit Title: Unit 2: Describing and Displaying Data** **Grade: 12**

| Lessons | Learning targets | Assessment tools and strategies<br>Formative/ Summative/ Differentiation | Resources and links for Parents |
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| **Lesson 1: Stem-and-Leaf Plots and Misleading Graphs** | • I can construct and interpret a stem-and-leaf plot.<br>• I can identify and explain features that make a graph misleading. | 1.2 Alternatives for auditory information (captions, transcripts, visuals).<br>8.3 Foster collaboration and community with structured group roles and supports.<br>Choose an item. | Links to support learning:<br>How to interpret a histogram \| Data and statistics \| 6th grade \| Khan Academy<br>Bar Charts, Pie Charts, Histograms, Stemplots, Timeplots (1.2)<br>Measure Central Tendency : Mean-Median-Mode and Range |
| **Lesson 2: Written Formative Assessment** | Learning targets:<br>• I can construct a categorical frequency distribution.<br>• I can construct a grouped frequency distribution.<br>• I can construct a histogram.<br>• I can recognize common distribution shapes: bell-shaped, positively/right-skewed, negatively/left-skewed, J-shaped, reverse J-shaped, bimodal, and U-shaped.<br>• I can construct a stem-and-leaf plot. | | |
| **Lesson 3: Population Variance and Standard Deviation** | • I can calculate and interpret population variance and standard deviation using the correct notation. | | |

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| **Lesson 4: Sample Variance and Standard Deviation** | • I can calculate sample variance and standard deviation and use them to compare variability in datasets. | | |
| **Homework** | • Complete the tasks in your Notebook | 8.1 Make goals clear, visible, and supported by short-term objectives. Choose an item. | |
| **Teacher's notes/ Reminders** | • Check **Microsoft Teams regularly** for class posts, announcements, resources, and assignments. | | |

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| **Technology Integration** | Microsoft (Power point, excel, etc.) |
| **Cross-curricular links** | **English / Media Literacy:** Students interpret survey questions, headlines, and data-based claims, then communicate and justify conclusions using clear evidence. |
| **External Assessment Integration** | SAT |
| **Wellbeing/ ATL Skills** | **Thinking Skills – Critical Thinking:** Students analyse data sources, identify bias, classify information, and evaluate whether conclusions are supported by the data.<br>**Communication Skills:** Students discuss findings, justify choices, and use appropriate statistical vocabulary.<br>**Research Skills:** Students plan data collection and evaluate the reliability and relevance of information. |
| **Standards** | • **HS.S-ID.A.2** Use statistics appropriate to the shape of the data distribution to compare center (median, mean) and spread (interquartile range, standard deviation) of two or more different data sets.<br>• **HS.S-ID.A.3** Interpret differences in shape, center, and spread in the context of the data sets, accounting for possible effects of extreme data points (outliers). |
| **Inquiry/ TOK Question** | **How can we collect and evaluate data to make reliable conclusions about a population?** |
| **Success Criteria (Measurable)** | **Lesson 1:**<br>• **Must:** Split values into stems and leaves, include a key, and identify a graph's axes and scale. |

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| | • **Should:** Place every value in the correct row, order the leaves, read values and clusters from the plot, and explain at least two ways a graph could mislead a reader.<br>• **Could:** Correct a misleading graph and explain how the revision changes a reader's interpretation while preserving the underlying data.<br>**Lesson 2:**<br>• **Must:** All students must be able to read frequency tables and data displays to identify frequencies and recognize common distribution shapes.<br>• **Should:** Most students should be able to organize data into categorical and grouped frequency distributions and construct accurate histograms and stem-and-leaf plots.<br>• **Could:** Some students could be able to interpret and compare data distributions, justify their conclusions using features of the displays, construct and verify a histogram.<br>**Lesson 3:**<br>• **Must:** I must be able to calculate population variance and standard deviation, showing my working.<br>• **Should:** I should be able to explain what standard deviation reveals about the spread of a population's data.<br>• **Could:** I could justify why variance has squared units and standard deviation has the same units as the original data.<br>**Lesson 4:**<br>• **Must:** I must be able to calculate sample variance and standard deviation, dividing the sum of squared deviations by one less than the sample size.<br>• **Should:** I should be able to compare two datasets using their sample standard deviations and explain which has greater variability.<br>• **Could:** I could evaluate a claim about stock-price variability using calculated standard deviations and evidence from the graphs. |
| **National Identity Domain** | Volunteering التطوع Global Understanding التفاهم العالمي Heritage التراث<br>Area: Community Responsibility and Informed Decision-Making |

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| | Implementation: Calculate the variance and standard deviation of daily water consumption for every household in a small UAE community over a specified day. Discuss how understanding variation can support responsible water use and sustainability. |